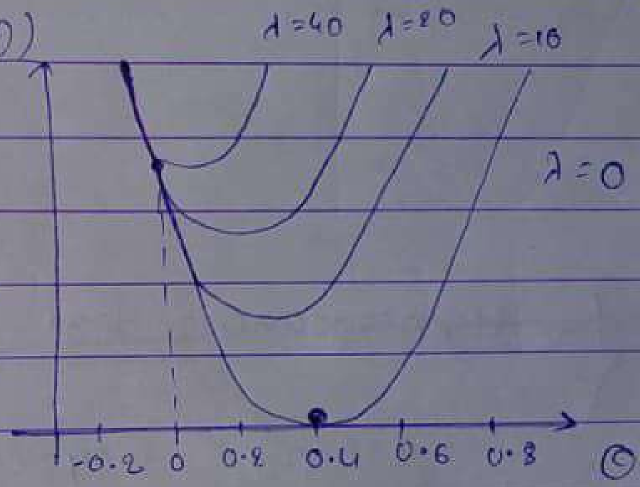
(2) Lasso Regression (L_1 Regularization)

↓

Feature selection

Cost fn :- $\frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x)^{(i)} - y^{(i)} \right)^2 + \boxed{\lambda \sum_{i=1}^{n} |slope|}$

$h_\theta(x) =$



$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$

$h_\theta(x) = 0.52 + 0.65 x_1 + 0.72 x_2 + 0.34 x_3 + \boxed{0.12} x_4$

⇓

Lasso Regression

$= 0.52 + 0.51 x_1 + 0.60 x_2 + 0.14 x_3 + \boxed{0 \times x_4}$

Note: It is used for feature selection, feature which coefficient is zero then remove this feature.

(3) Elastic Net Regression

- → (1) Reduce Overfitting
  → (2) feature selection

$$\text{cost fn} = \frac{1}{2m}\sum_{i=1}^{m}\left(h_\theta(x)^{(i)} - y^{(i)}\right)^2 + \underbrace{\boxed{\lambda_1 \sum_{i=1}^{m}(\text{slope})^2}}_{\text{Reduce overfitting}} + \underbrace{\boxed{\lambda_2 \sum_{i=1}^{m}|\text{slope}|}}_{\text{feature selection}}$$

{ Hyperparameter Tuning the Linear Regression }